Beta-blocker side-effects in clinical practice: A nationwide approach

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ABSTRACT

Background Concerns regarding side-effects of beta-blockers (BBs) are frequent but data regarding the incidence of side-effects are conflicting and real-world data are sparse. Hence, we aimed to investigate the absolute and relative risks of BB side-effects in clinical practice.

Methods Using Danish nationwide registers, we included Danish hypertensive patients initiating antihypertensive treatment with a BB or calcium-channel blocker (CCB). We computed crude as well as standardized 1-year risks and adjusted risk ratios of BB side-effects (depression, anxiety/insomnia, gastrointestinal side-effects, erectile dysfunction, and dizziness/fainting) compared with CCB treatment.

Results We included 64,722 patients initiating treatment with a BB and 181,880 patients initiating treatment with a CCB. In patients initiated on BB, the standardized 1-year risk of any outcome, erectile dysfunction exempt, was 13.7% (95% CI: 13.4%-13.9%). The 1-year risk of specific BB side-effects was the highest for anxiety/insomnia (6.2%, 95% CI: 6.0%-6.3%), gastrointestinal side-effects (4.6%, 95% CI: 4.4%-4.7%), and erectile dysfunction (4.7%, 95% CI: 4.5%-4.9%). The risk of side-effects was consistently increased when comparing BB treatment with CCBs including depression (Risk Ratio [RR] 1.48, 95% CI 1.41-1.55), anxiety/insomnia (RR 1.53, 95% CI 1.47-1.59), gastrointestinal side-effects (1.31, 95% CI 1.25-1.36), and dizziness/fainting (RR 1.50, 95% CI 1.38-1.61), but not erectile dysfunction (RR 0.91, 95% CI, 0.85-0.96).

Conclusions In a large nationwide cohort, the incidence of BB side-effects was clinically relevant and consistently increased compared with CCBs with the exception of erectile dysfunction, which carried similar risks for both treatments. (Am Heart J 2026;291:111–120.)

Background

Beta blockers (BBs) are a commonly prescribed class of cardiovascular drugs used for antagonizing beta-adrenergic receptors, which are key in regulating several autonomic physiological processes in the human body (eg, blood pressure, heart rate, metabolic- and nervous system processes).[1] Consequently, BBs have a variety of clinical indications pertaining to the cardiovascular system with documented beneficial effects in conditions such as congestive heart failure, tachy-arrhythmias, hypertension, as well as ischemic heart disease.[2-5]

In relation to the indication to treatment, different generations of BBs differ in their selectivity (cardioselectivity vs. nonselectivity) as well as vasodilatory properties.[6] Since beta-adrenergic receptors are widely distributed in the body, BB side-effects are diverse and relate to several different organ systems (eg, depression, dizziness and orthostatic hypotension, erectile dysfunction, anxiety and insomnia, and metabolic side-effects).[7-10] However, data from randomized trials have provided indications that the described side-effects are not necessarily more frequent in patients treated with BBs than placebo.[11] Moreover, it has been documented that meticulously informing patients of a potential side-effect (eg, erectile dysfunction) could induce a 'nocebo-effect' as the information in itself confers an increased probability of experiencing the given side-effect.[12]

However, there is a paucity of reliable data outside the confinements of clinical trials regarding the incidence of

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side-effects in patients treated with BBs in clinical practice providing basis for informing patients. Consequently, using Danish nationwide registers, we sought to investigate the incidence of BB side-effects in patients with hypertension compared with calcium-channel blockers (CCBs).

Methods

Data sources

In Denmark, a network of administrative and clinical registers and databases with nationwide coverage enables large-scale epidemiological research. Using a unique identifier (ie, the CPR number) all residents can be followed on an individual level linking different databases pertaining to demographic variables (ie, age, sex, death, education level), hospital admissions and diagnoses (International Classification of Diseases 10th edition [ICD-10]), hospital procedures and operations, as well as medicine dispensed at any pharmacy in Denmark.[13-15] The registers have been described elsewhere in more detail and have been used extensively and validated, especially for cardiovascular research.[16-18] Enrollment in the databases is mandatory for all citizens in Denmark and does not require individual consent.

Study design, population, and exposure

The study was conducted as a nationwide cohort study including all patients ≥40 years of age initiating treatment for hypertension with either a BB (metoprolol, bisoprolol, carvedilol, atenolol, propranolol, and other [<1%]) or a CCB (amlodipine, lercanidipine, verapamil, and other [<1%]) between 2012 and 2023. Patients were required to be naïve to the mentioned drugs. Hypertension was defined as treatment with at least 2 types of antihypertensive medication as BBs are no longer a first-line therapy for hypertension.[19] As such, all patients were in antihypertensive treatment with at least 1 other drug prior to BB or CCB treatment initiation. Using claimed prescriptions for defining hypertension in the Danish registers is a method that has been previously used and validated.[20] Patients with other potential indications for BB- and CCB treatment (eg, heart failure, tachyarrhythmias, ischemic heart disease, hyperthyroidism, and essential tremor) were excluded to obtain a homogenous study population and appropriate comparator groups. Likewise, patients with medical history of any of the outcomes were excluded as well. (Figure 1) Codes (eg, ICD-10 and ATC codes) used for defining variables for inclusion and exclusion criteria, please see Supplementary Table S1.

Outcomes and follow-up

Five different outcomes identified as markers of potential quality-of-life reducing side-effects to BB treatment were predetermined as: (1) depression, (2) erectile dysfunction, (3) anxiety or insomnia, (4) gastrointestinal side-effects, (5) dizziness, fainting or orthostatic hypotension. (1) Depression was defined as a claimed prescription of an antidepressant; (2) erectile dysfunction by a claim of a prescription for phosphodiesterase type 5 inhibitor; (3) anxiety or insomnia by a claimed prescription of anxiolytics or sedating medication used for treating insomnia; (4) gastrointestinal side-effects by diagnostic codes for constipation, or a claimed prescription of a laxative; (5) dizziness or fainting by diagnostic codes for dizziness, fainting, collapse, or orthostatic hypotension. "Any outcome" was defined as well as a composite of all described outcomes except erectile dysfunction as only male patients are included in the analysis of this specific outcome. Specific ICD-10 and ATC codes are listed in Supplementary Table S1.

Patients were followed from the date of their first-time claim of either a BB or a CCB until event of interest, emigration, death, end of study period (31 December 2023), or 1 year of follow-up. Specifically for patients initiated on CCB, follow-up was also halted if a prescription of a BB was claimed during follow-up since this would place these patients "at risk" in this active comparator design. On the contrary as CCB treatment was assumed not to be associated with the outcomes, a claimed prescription of CCB in the BB group during follow-up was ignored.[21]

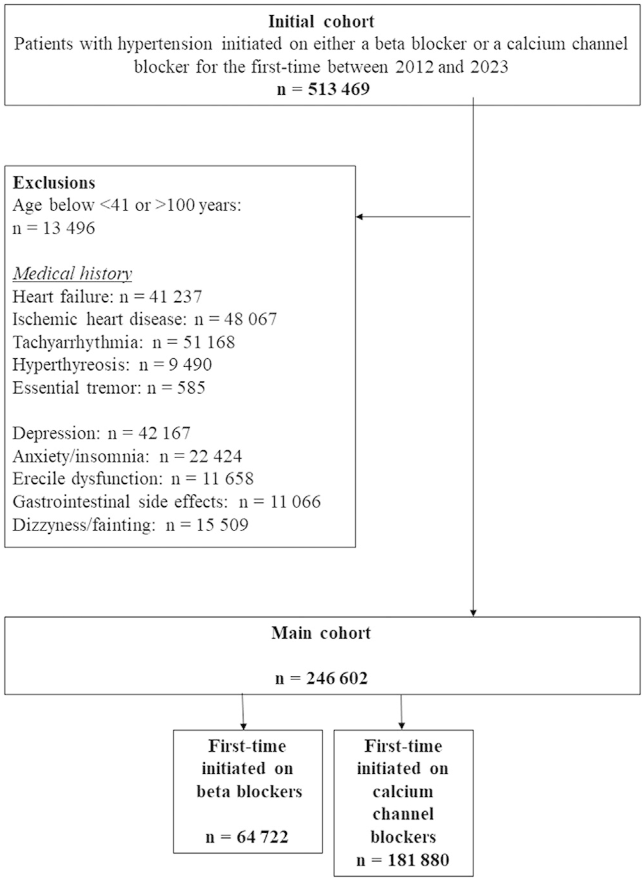
Concomitant pharmacotherapy and comorbidity

Baseline characteristics were defined at index-date (initiation of treatment with either a BB or CCB). Comorbidity included in the study was a medical history of chronic obstructive pulmonary disease, stroke, chronic kidney disease, peripheral artery disease, and cancer. (Supplementary Table S1). Comorbidities were captured using diagnostic codes (ICD-10) in a period of 5 years prior to study inclusion as done previously. Medical treatment considered at baseline was treatment with renin-angiotensin-system inhibitors, thiazides, loop diuretics, statins, antiplatelets, anticoagulants, nonsteroid anti-inflammatory drugs, opioids, and glucose lowering drugs (used as a proxy for diabetes mellitus). Educational level was categorized as elementary- or high school, vocational education, higher education, or unknown.

Statistical analysis

Characteristics in categories at baseline were shown as both total counts and percentages within each group of interest. Age was presented as median with interquartile range (IQR). To describe the absolute risk of possible BB side effects, the Aalen-Johansen estimator was used to calculate crude 1-year absolute risks with 95% confidence intervals (CI) treating death as competing risk.[22] Using an active comparator design, where BB treatment was compared with CCB treatment, standardized absolute 1-year risks following initiation of both

Figure 1. Flowchart. Flowchart showing the application of exclusion and inclusion criteria into the final cohort.



drugs and risk ratios were computed.[21] First, a multivariable, cause-specific cox model was fitted for the specific outcomes including the following predetermined variables: age, sex, educational level, and medical history of stroke, diabetes mellitus, chronic obstructive pulmonary disease, cancer, and chronic kidney disease. Second, the G-formula was executed to compute average (adjusted) treatment effects and provide adjusted absolute risks and corresponding risk ratios (RR) comparing BB treatment with CCB treatment.[23,24]

Several supplementary analyses were performed: 1) analyses were performed divided into two age strata (>65 years and ≤ 65 years of age) examining different presentations of BB side-effects according to age, 2) analyses were performed stratified by the specific BB/CCB subtype, 3) analyses were performed stratified by sex (erectile dysfunction not included), 4) and an analysis investigating a potential dose-response relationship was performed. The category "high-dose" was defined as patients claiming prescriptions for the highest tablet strength available for the different BBs (≥100mg metoprolol, ≥10mg bisoprolol, ≥25mg carvedilol, ≥100mg atenolol, ≥80mg propranolol). Patients claiming the "high dose" were then compared with patients receiving other and lower strengths of the drugs categorized as "low dose". As a marker of treatment adherence during follow-up, we descriptively assessed the proportion of patients claiming a second prescription after treatment initiation within a year using the Aalen-Johansen estimator with death as a competing risk.[22] A similar approach was applied to show the proportion of patients claiming a third prescription following a second. Further, to investigate the different outcomes' potential impact on adherence, the proportion of claiming another prescription was calculated in a similar manner among patients experiencing an outcome. Ultimately, an analysis examining time trends (ie, stratifying analyses before and after 2017) as well as analyses by frailty score were performed.[25] All data management, figures, and statistical analyses were conducted using R version 4.4.1.[26]

Ethics

Retrospective, observational studies based on administrative data do no not require ethical approval in Denmark. Permission was granted to the institution by the responsible government data institute and the project has been registered.

Funding

No extramural funding was used to support this work. The authors are solely responsible for the design and conduct of this study, all study analyses, the drafting and editing of the paper and its final contents.

Results

A total of 64,722 patients initiating treatment with a BB and 181,880 patients initiating treatment with CCBs were identified and included in the final cohort. (Figure 1) During the study period from 2012 to 2023, a switch toward fewer patients being initiated on treatment with a BB could be observed compared with CCBs. (Supplementary figure S1)

In general, in both treatment groups (BBs and CCBs) men and women were equally represented, albeit the BB group was marginally older compared with CCBs (Median age 67 years vs 64 years). Between the BB and CCB treated patients, we observed almost a comparable prevalence of diabetes (17.7% vs 15.9%), stroke (7.7% vs 6.8%)), and cancer (10.9% vs 9.2%). However, there were slightly more patients with a registered diagnosis of kidney disease in the BB treated group compared with CCBs (4.7% vs 2.7%). (Table 1)

In the BB group, the majority was treated with metoprolol (76.2%), and to a lesser extent propranolol (7.0%), carvedilol (6.1%), and bisoprolol (6.1%). The predominant CCBs used in the cohort were amlodipine (84.1%) and lercanidipine (11.4%). More patients treated with a BB were in treatment with >2 types of antihypertensive medication than in the CCB group (59.2% vs 38.5%). By far the most prevalent concomitant antihypertensive drug treatment was with a renin-angiotensin-system inhibitor. (Table 1). Adherence was slightly higher in the CCB treated group, as the cumulative incidences of claiming a second prescription within the first year were 81.1% (95%CI: 80.8%-81.4%) and 85.9% (95% CI: 85.7%-86.1%) and slightly higher for a third prescription, respectively for BB- and CCB treatment. (Supplementary Figure S2) Moreover, a lower adherence to both BB and CCB treatment could be observed among patients experiencing a side-effect. (Supplementary Table S2)

The overall crude absolute risk of experiencing at least 1 side-effect—erectile dysfunction *not* included—within 1 year of treatment with BB treatment was roughly 1 in 7 (one-year risk: 14.4%, 95% CI 14.2%-14.7%) with a median time to first outcome of 122 days (interquartile range 44-230), which was higher than in the CCB group (one-year risk: 9.6%, 95% CI 9.4%-9.7%). Median time to first outcome in the CCB group was 131 days (interquartile range 53-238).

The crude risk of BB specific side-effects within 1 year after treatment initiation was the highest for anxiety/insomnia (one-year risk 6.4%, 95% CI 6.2%-6.6%), gastrointestinal side-effects (one-year risk 5.1%, 95% CI 5.0%-5.3%), and depression (one-year risk 4.2%, 95% CI 4.1%−4.4%) as well as erectile dysfunction (one-year risk 4.6%, 95% 4.4%-4.8%), and to a lesser extent dizziness/fainting (one-year risk 1.8%, 95% CI 1.7%-1.9%). (Figure 2) The risks of the same side-effects among patients initiated on CCBs were lower, except a comparable risk found for erectile dysfunction in men. (Supplementary Figure S3)

Using multivariable cox models computing standardized absolute risks comparing BB side-effects with CCBs revealed increased risks associated with BBs for all side-effects except erectile dysfunction. (Graphical abstract) Specifically, we found increased standardized RRs for anxiety/insomnia (RR 1.53, 95% CI 1.47-1.59), depression (1.48, 95% CI 1.41-1.55), gastrointestinal side-effects (1.31, 95% CI 1.25-1.36), and dizziness/fainting (RR 1.50, 95% CI 1.38-1.61). In men, the risk of erectile dysfunction was not higher in patients treated with BB compared with CCB (RR 0.91, 95% CI, 0.85-0.96). (Figure 3)

A stepwise association between higher doses of BBs and higher risk of side-effects compared with CCBs was found, again except for erectile dysfunction where both low and high dosage of BB seemed comparable to CCB. (Figure 4)

Supplementary analyses

Separating analyses in two age groups (≤65 years & >65 years) revealed that erectile dysfunction, and depression were the highest among youngest patients (Supplementary Figure S4A). Contrarily, in older patients, BBs conferred higher risks of gastrointestinal side-effects and dizziness/fainting (Supplementary Figure S4B).

Sex-specific analyses showed that dizziness/fainting and gastrointestinal side-effects were equally common between the sexes, but depression and anxiety/insomnia was higher among women compared with men (Supplementary Figure S5A and S5B).

Additional analyses stratifying patients by the number of antihypertensive drugs in the treatment regimen (>2 drugs vs 2 drugs) did not affect the results (Supplementary Figure S6A and S6B).

Analyzing the BBs individually revealed notable differences with the highest risk of depression, anxiety/insomnia, and gastrointestinal side-effects associated with propranolol whereas bisoprolol and carvedilol conferred the highest risks of erectile dysfunction and dizziness/fainting, respectively. (Supplementary Table S3A)

Stratifying the main analyses by calendar year revealed slightly higher absolute risks of the outcomes during 2012-2017 compared with 2018-2023, but similar relative risks were found when comparing BB treatment with CCB treatment. (Supplementary Figure S7A and S7B)

Table 1. Baseline characteristics

	Calcium channel blockers	Beta blockers
N	181,880	64,722
Women, n (%)	86,875 (47.8)	32,791 (50.7)
Age in years, median [interquartile range]	64 [55, 72]	67 [58, 75]
More than 2 antihypertensive agents, n (%)	69,989 (38.5)	38,335 (59.2)
Beta blocker agent		
Metoprolol	-	49,320 (76.2)
Propranolol	-	4,501 (7.0)
Carvedilol	-	3,928 (6.1)
Bisoprolol	-	3,952 (6.1)
Atenolol	-	982 (1.5)
Other	-	2,039 (3.2)
Calcium channel blocker agent		
Amlodipine	152,961 (84.1)	-
Lercanidipine	20,652 (11.4)	-
Felodipine	6,195 (3.4)	-
Nifedipine	935 (0.5)	-
Verapamil	873 (0.5)	-
Other	264 (0.1)	-
Comorbidity, n (%)		
Diabetes mellitus	28,866 (15.9)	11,457 (17.7)
Stroke	12,400 (6.8)	5,001 (7.7)
Cancer	16,650 (9.2)	7,066 (10.9)
Chronic obstructive pulmonary disease	4,747 (2.6)	2,486 (3.8)
Chronic kidney disease	4,903 (2.7)	3,027 (4.7)
Peripheral artery disease	2,060 (1.1)	1,279 (2.0)
Concomitant medication, n (%)		
Renin-angiotensin-system inhibitors	168,342 (92.6)	51,722 (79.9)
Thiazides	35,303 (19.4)	16,210 (25.0)
Loop diuretics	6,566 (3.6)	8,153 (12.6)
Antiplatelet agents	25,832 (14.2)	15,826 (24.5)
Statins	62,095 (34.1)	24,659 (38.1)
Nonsteroid anti-inflammatory drugs	28,400 (15.6)	10,148 (15.7)
Opioids	12,175 (6.7)	5,953 (9.2)
Educational level, n (%)		
Elementary- or high school	51,349 (28.2)	21,330 (33.0)
Vocational education	80,846 (44.5)	27,146 (41.9)
Higher education	46,186 (25.4)	14,453 (22.3)
Unknown	3,499 (1.9)	1,793 (2.8)
Clinical frailty score		
Low	164,179 (90)	57,004 (88)
Medium	17,253 (10	7,436 (12)
High	448 (0)	282 (0)

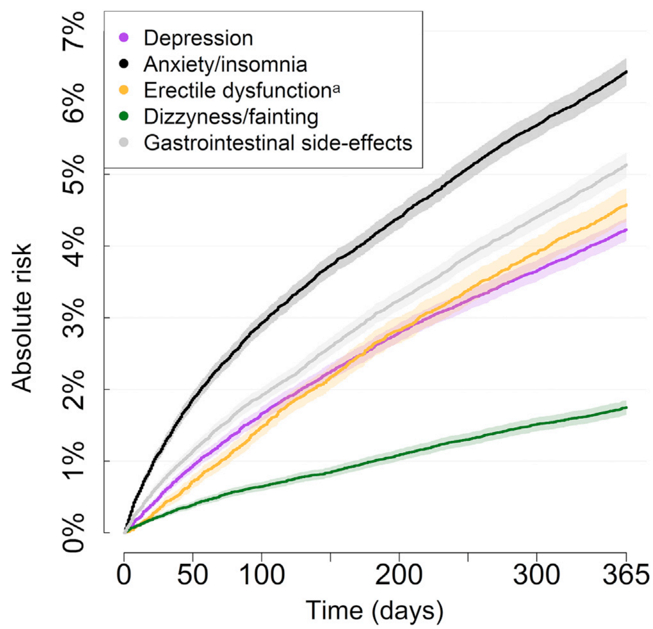
Likewise, if the analyses were stratified by frailty, similar relative risks were found, but patients in the "medium" or "high" frailty groups had higher risks of experiencing an outcome during follow-up. (Supplementary Figure S8A and S8B)

Discussion

In this nationwide cohort study investigating the risks of side-effects in patients with hypertension treated with BBs, more than one in ten patients experienced at least one of the investigated side-effects within a year of treatment initiation. Compared with CCBs, we uncovered significantly increased risks of important quality-of-life reducing side-effects including depression, anxiety/insomnia, gastrointestinal side-effects, and dizziness/fainting associated with BBs. Interestingly, the incidence of erectile dysfunction was not higher in men treated with BBs compared with CCB, despite it being a common concern regarding BBs. (Graphical abstract)

Numerous clinical trials have investigated the role of BBs in hypertension and found reductions in cardiovascular disease associated with BB treatment. However, BBs are likely inferior to other antihypertensive drugs (eg, CCBs, diuretics, renin-angiotensin system inhibitors) in terms of mortality reduction and protection against stroke.[27,28] Conversely, BBs are no longer a first-choice treatment of arterial hypertension; however, BBs are still

Figure 2. Crude risk of side-effects following beta blocker initiation. One-year absolute risk calculated for each outcome using the Aalen Johansen estimator taking censoring and competing risk of death into account. [a]Women were excluded from these analyses.



a central part of the treatment recommendations and algorithms in guidelines.[5] Moreover, despite often being a somewhat 'unpopular' group of drugs, BBs have documented cardioprotective effects, especially, in concert with other antihypertensive drugs or in the case of more than one indication for treatment (eg, atrial fibrillation and hypertension).[29]

Importantly, concerns regarding tolerability and potential side-effects are prevalent and studies clearly indicate that BBs are perceived by physicians as a class of drugs with general poor tolerability, a significant risk of side-effects, and inferior adherence.[30-32] However, meta-analyses of clinical trials including thousands of patients showed no effect on depressive symptoms as well as only a very small increase in risk of fatigue and erectile dysfunction and an inconsistent signal towards an increased probability of treatment discontinuation with BBs.[10,27] Notably, it seemed that side-effects were more common in the early generation BBs (eg, propranolol). This tendency was also, at least partly, represented in our data. Interestingly, subsequent analyses of trial data have indicated that the incidence of side-effects are not necessarily higher in BB treated patients than with placebo.[11] Moreover, an interesting trial showed that meticulously informing patients of a potential risk of erectile dysfunction prior to BB initiation dramatically increased the risk of experiencing the side-effect comparing to more sparse patient information.[12]

The pharmacological effects of BBs are mediated by antagonizing beta-adrenergic receptors in the sympathetic nervous system by specifically targeting beta-1 receptors (selective beta-blockade) or both beta-1 and beta-2 receptors (nonselective beta-blockade). Moreover, some newer BBs possess vasodilatory properties due to alpha-1 adrenergic receptor antagonism as well as increased release of endothelium-derived nitric oxide.[6] As such, tolerability and the incidence of side-effects could theoretically vary between different generations of BBs. In our data, differences in the risks of side-effects varied between the individual drugs, however, potential secondary and undetected indications for choosing a specific BB other than only hypertension (eg, propranolol against tremor) make direct comparisons difficult. Moreover, the majority of trial data regarding hypertension are based on the use of atenolol (~75%) and how these results extrapolate to other generations of BBs are unknown.[27]

In our data, the risk of experiencing at least 1 of the investigated side-effects—erectile dysfunction not included—within 1 year was moderate (~14%); however, it is important to note that we likely only capture the most severe presentations leading to hospital contact or requiring pharmacological treatment. As such, the real incidence of perceived or experienced side-effects could be higher.

Importantly, we found consistently increased risks of side-effects, including a dose-response relationship, compared with CCBs—surprisingly, not including erectile dysfunction where a comparable to slightly lower risk was found. This is noticeable since erectile dysfunc-

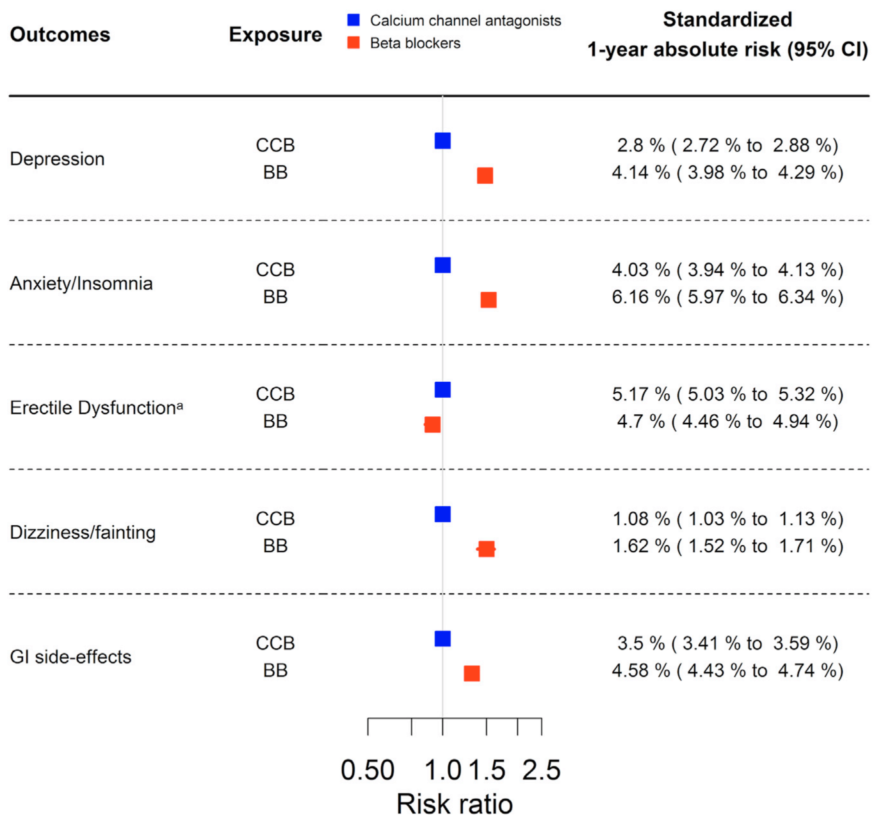


Figure 3. Associations between beta blockers and side-effects compared with calcium channel blockers. Forest plot depicting standardized or adjusted 1-year absolute risk and corresponding risk ratios comparing beta blocker with calcium channel blocker treatment. These analyses were based on cause-specific cox models adjusted for predetermined baseline variables: sex; age; educational level; and medical history of diabetes mellitus, cancer, chronic kidney disease, stroke, and chronic obstructive pulmonary. [a]Women were excluded from these analyses.

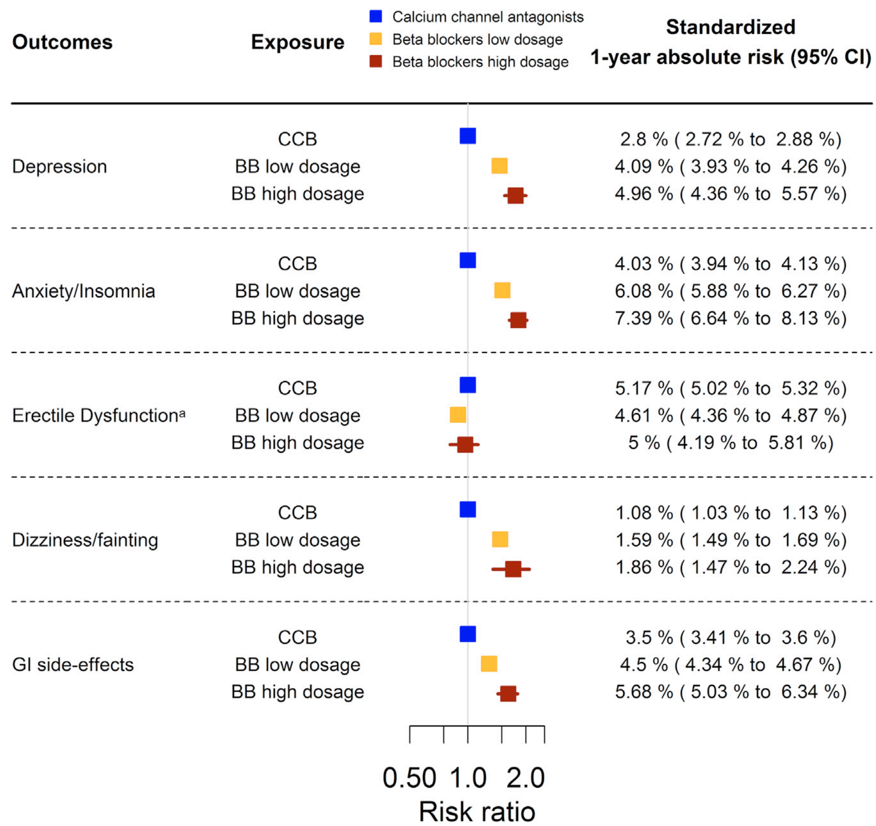
tion is a common concern regarding beta-blockers. Nevertheless, regarding depression, anxiety/insomnia, gastrointestinal side-effects, and dizziness/fainting, our data clearly indicate that real-world incidence of side-effects are higher in BB treated patients than with CCB with potential impact on perceived patient quality-of-life. This was also reflected by more patients on CCB claiming a second prescription within the first year (86% vs 81%) and a higher rate of BB discontinuation among patients experiencing a side-effect than the remaining BB treated patients—a trend also present among patients initiated on CCBs.

However, it is important to note that real-world patients will often be informed of potential side-effects of BBs before treatment initiation and will be susceptible to the 'nocebo'-effect mediated by fear or expectation of a potential side-effect.

Moreover, the nature of the side-effects seems to be dependent on age as younger patients are more prone to being bothered by erectile dysfunction and side-effects related to the central nervous system (depression) than older patients where gastrointestinal side-effects, dizziness, and fainting seem more dominant. Likewise, it is also worth noting, that patients with higher clinical frailty scores had higher risks of most outcomes compared with lower-scoring patients potentially indicating a patient group of improved vigilance. However, this increased risk could be associated with the patients' general health status as the relative risk differences between the BB and CCB group seemed comparable to the main analyses. The higher incidence of depression and anxiety/insomnia side-effects in women is conspicuous and should warrant further investigation.

Conclusively, our study underlines the fact that in real-world clinical practice, perceived side-effects to BB treatment are relatively frequent and likely higher than in patients treated with other antihypertensive drugs. Additionally, the risk seemed accentuated with an increasing dose of BBs.

Figure 4. Standardized 1-year absolute risk and risk ratios comparing low- and high dosage (≥100mg metoprolol, ≥10mg bisoprolol, ≥25mg carvedilol, ≥100mg atenolol, ≥80mg propranolol) of beta blockers with calcium channel blocker treatment were calculated using the G-formula to gain average treatment effects. These analyses were based on cause-specific cox models adjusted for sex; age; educational level; and medical history of diabetes mellitus, cancer, chronic kidney disease, stroke, and chronic obstructive pulmonary. [a]Women were excluded from these analyses.



Limitations

Our study has several important limitations that should be addressed. Importantly, our data is based on the clinical and administrative registers and databases in Denmark. Many of the registered diagnoses and the data used have been leveraged for research purposes for decades and validated extensively. However, the risk of misclassification bias cannot be ruled out as not all diagnoses and data have been manually evaluated and validated.[13,14,33] Importantly, numerous validation studies have been performed generally revealing high data quality, especially regarding cardiovascular diagnoses.[16] As previously mentioned, patients had to either receive pharmacological treatment for a side-effect or require hospital contact to be captured as a side-effect in our study. As such, our absolute risks of side-effects are potentially somewhat underestimated; however, relative risks would be unaffected by this. Choosing appropriate comparator groups is of importance and was considered in detail as BBs are no longer a first-line choice for antihypertensive treatment. To account for this, all patients had to receive the index drug prescription (BB or CCB) as an added additional drug for hypertension. Moreover, we used state-of-the art causal inference statistical techniques to account for residual confounding. However, it cannot be completely ruled out that patients included in the BB group could be different, and at least a part of the increased risks of side-effects were inherent to their patient characteristics and not the treatment. It should also be noted that the patients in our cohort are likely not the least complex hypertensive patients, as they are all in treatment with at least 2 different antihypertensive drugs. Moreover, how our results apply for other groups of patients (eg, atrial fibrillation) is unknown, however, we have no reason to expect major differences.

Conclusion

In a large nationwide cohort, more than one in ten patients initiating treatment with BBs experience at least one significant side-effect including depression, anxiety/insomnia, gastrointestinal side-effects, and dizziness/fainting. The risk of all side-effects was increased when comparing treatment with CCBs in hypertension with the exception of erectile dysfunction in which an increased risk associated with BBs could not be observed. Albeit a certain "nocebo"- effect could be present as physicians are likely vigilant in terms of informing patients of potential side-effects, our data clearly indicate that BB side-effects remain a prevalent issue in pharmacological treatment and prevention of cardiovascular disease.

Data availability

Danish registry data cannot be made publicly available by the authors, as data access to must be granted to research institutions on an individual basis by the Danish Data Protection Agency and Statistics Denmark. For questions pertaining to data access inquiries should be addressed to Statistics Denmark (https://www.dst.dk/da/TilSalg/Forskningsservice/Dataadgang).

Declaration of competing interest

None reported.

CRediT authorship contribution statement

Peter Vibe Rasmussen: Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Jarl Emanuel Strange:** Writing – review & editing, Writing – original draft, Methodology. **Sebastian Kinnberg Nielsen:** Writing – review & editing. **Rasmus Borup Hansen:** Writing – review & editing. **Gunnar H. Gislason:** Writing – review & editing, Data curation. **Morten Lamberts:** Writing – review & editing. **Morten Lock Hansen:** Writing – review & editing, Conceptualization. **Anders Holt:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ahj.2025.08.005.

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